

Commonly Used Performance Metrics for Evaluating Vehicular Communication Systems

Successful Message Ratio (SMR)

Definition:

The *Successful Message Ratio (SMR)* is the fraction of vehicles within the communication range of a sender that successfully receive its broadcast message. It is a measure of the **reliability** of message delivery in vehicular networks.

Formula:

$$SMR(\%) = (N_{recv} \div N_{total}) \times 100$$

- **Where:**

N_{recv} = Number of vehicles that successfully received the broadcast message.

N_{total} = Total number of vehicles within the sender's communication range (potential receivers).

- **Example:** If a sender has **10 neighboring vehicles** within range, and **7 of them successfully receive** the message: $\rightarrow SMR = 7/10 = 0.7$ SMR = 70%

Average Successful Message Ratio (SMR_{avg})

Definition:

The *Average SMR* is the mean reliability of broadcast message delivery across multiple transmissions or multiple senders. It represents the **average fraction of successfully received messages** by potential receivers in the network.

$$\circ \quad SMR_{avg} = \frac{\sum_{j=1}^M N_{recv}(j)}{\sum_{j=1}^M N_{total}(j)}$$

Where

M = Total number of broadcast transmissions (or sender-message pairs) considered.

$N_{recv}(j)$ = Number of vehicles that successfully received the j – th transmission.

$N_{total}(j)$ = Total number of vehicles within the sender's communication range for the j – th transmission.

Important Note on SMR and SMR_{avg} :

In practical wireless networks, **neither the transmitting node nor the receiving nodes can calculate the Successful Message Ratio (SMR) or the Average SMR (SMR_{avg}) on their own** for broadcast frames (e.g., Basic Safety Messages, BSMs). This is because:

- Broadcast messages in IEEE 802.11/802.11p do not use acknowledgments (ACKs).
- The transmitter has no feedback about which neighbors successfully received the packet and which did not.
- Similarly, each receiver only knows what it successfully received, but not the total set of potential receivers or how many of them also received it.

As a result, **only an external observer with global knowledge of the network traffic** (e.g., a network administrator or monitoring system) can accurately compute SMR/ SMR_{avg} , since this requires:

1. Knowing the **total number of potential receivers** within communication range for each transmission.
2. Knowing the **actual number of nodes that successfully received** each broadcast.

In real-world deployments, this type of measurement is done through **network-wide monitoring, logging, or controlled experiments** rather than by individual nodes.

In our case, since we are working with simulations (e.g., in **Veins** with OMNeT++ and SUMO), the simulator inherently has access to all transmission and reception events across the network. Therefore, we can calculate SMR and SMR_{avg} directly from the simulation logs, even though this would not be possible for a single node in a live system.

No Message Interval (NoM)

Definition:

The *No Message Interval (NoM)* quantifies the **longest continuous time period** during which a receiver does not successfully receive any messages from a given sender. It highlights the worst-case gap in communication (communication outage), making it a key reliability and timeliness metric in vehicular networks.

Interpretation:

- While *Inter-Reception Time (IRT)* reports the gap between every consecutive reception, *NoM* focuses **only on the longest gap**.
- A high NoM value indicates a risk of missing critical updates for an extended period.

Example:

If a vehicle receives messages at times 1.0s, 1.1s, and 1.6s, the gaps are 0.1s and 0.5s.

Thus: NoM=0.5 s

Relevance in VANETs:

- Critical for **safety applications**, where long gaps without updates may compromise awareness.
- Useful for evaluating **worst-case performance** in high mobility or congested channel conditions.

End-to-End Delay (E2E)

- **Definition:** Time between **message generation** at the sender and **reception** at the receiver.

How to Calculate **E2E** using generated .CSV:

To calculate the **End-to-End (E2E) Delay** from the simulation results:

1. **Extract Delay Data (from `veinsAppLayerMessageInfo.csv`):**
 - The file `veinsAppLayerMessageInfo.csv` contains detailed records for each received broadcast message (e.g., BSM).
 - For every received packet, the file logs the end-to-end delay.

Channel Access Delay (CAD)

- **Definition:** Waiting time before a transmitter can access the channel (due to contention/backoff).

Important Note on E2E and CAD:

Understanding Delay Components

Delay Components

- **Channel Access Delay (CAD):** Waiting due to CSMA/CA backoff.
- **Transmission (TX):** Time to put bits on air.
- **Propagation:** Travel time over the channel (very small).
- **Retransmissions:** Extra CAD + TX + control frames (unicast only).

Broadcast vs. Unicast Delay Components

Broadcast (Safety Messages)

- **One attempt only:** No ACK, no retransmission.
- **High priority** (shorter backoff).
- **Delay timeline:**
 $E2E \approx CAD + TX + Prop$
- **Effect:** CAD appears once and is small.
- **Tradeoff:** If collision occurs, packet is lost (reliability drop, not extra delay).

Unicast (Non-Safety Data)

- **ACK required.** If no ACK → retransmission.
- **Each attempt repeats CAD + TX**, with larger backoff windows.
- **Delay timeline:**
 $E2E \approx \sum (CAD_i + TX_i) + ACK + \text{timeouts}$
- **Effect:** CAD becomes **dominant** because it repeats with every retry.

Comparison

Aspect	Broadcast (Safety)	Unicast (Non-Safety)
ACK / Retransmissions	No	Yes
CAD Occurrence	Once	Repeats (grows with load)
Reliability Strategy	Best effort	Guaranteed by ACK
Delay Impact	CAD minor	CAD significant
Result	Low delay, lower reliability	Higher delay, higher reliability

Summary:

- **Broadcast:** Fast, one-shot → CAD is not prominent in total delay.
- **Unicast:** Reliability via ACK/retransmissions → CAD accumulates, often dominates end-to-end delay.
- In short:
Broadcast = “Send once, fast” → CAD small.
Unicast = “Retry until success” → CAD grows and delays increase.

Metrics Insight

Metric	Indicates	Affected by
SMR_{avg} (Average Successful Message Ratio)	Reliability of broadcast message reception	- Packet collisions - Interference - Path loss/fading - Network density (number of neighbors)
NoM (No Message Interval)	Worst-case timeliness (longest gap between receptions, maximum communication outage)	- Channel congestion - Collisions - Interference - High mobility (rapid topology changes)
E2E Delay (End-to-End Delay)	Latency of message delivery (generation → reception)	- Queuing delay - Propagation delay - Channel contention - Retransmissions
CAD (Channel Access Delay)	Medium access efficiency (waiting time before transmission)	- Backoff mechanism - Contention window size - Channel load (number of competing nodes)

Relationship Between Metrics

- **SMR_{avg}** → reliability of reception.
- **NoM** → timeliness, with NoM showing worst-case gaps.
- **E2E Delay** → latency of message delivery.
- **CAD** → channel contention effects.

For **safety-critical VANETs**, desirable performance is:

- **High SMR_{avg}** (≈95%)
- **Low NoM** (timely updates)
- **Low E2E Delay** (fast delivery)
- **Low CAD** (quick channel access)